

AYJR 2025 (April) - Evening Shift By: C I P H E R

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1: (Only One Option Correct Type)

Let A be a square matrix of order 3 such that $\text{adj}(\text{adj}(\text{adj}(A))) = \begin{bmatrix} 16 & 0 & 4 \\ 5 & 4 & 0 \\ 1 & 4 & 3 \end{bmatrix}$ and $\det(A)$ is positive, then

which of the following must be correct?

- (A) $8 \cdot \text{trace}(A^{-1}) = 32$
- (B) $\det(\text{adj } A) = 4$
- (C) $8 \cdot \text{trace}(A^{-1}) = 23$
- (D) $\det(\text{adj } A) = 2$

Question No. 2: (Only One Option Correct Type)

Among the $8!$ permutations of the digits $1, 2, 3, \dots$, consider those arrangements which have the following property. If we take any five consecutive positions, the product of the digits in these positions is divisible by 5. The number of such arrangement is equal to

- (A) $7!$
- (B) $2 \cdot (7!)$
- (C) 7C_4
- (D) None of these

Question No. 3: (Only One Option Correct Type)

If $(x - 2 + \frac{1}{x})^{30} = n_0x^{30} + n_1x^{29} + \dots + n_{29}x + n_{30} + n_{31}x^{-1} + \dots + n_{60}x^{-30}$ and $C = n_0 + n_1 + n_2 + \dots + n_{60}$. Find the value of $(a + b)$ if $C - n_{30} = -\binom{a}{b}$. [Note: $\binom{n}{r}$ denotes nC_r .]

- (A) 30
- (B) 45
- (C) 60
- (D) 90

Question No. 4: (Only One Option Correct Type)

Let the foot of perpendicular from a point $P(1, 2, -1)$ to the straight line $L : \frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$ be N . Let a line be drawn from P perpendicular to vector $\hat{i} + \hat{j} + 2\hat{k}$ which meets L at point Q . If α is the acute angle between the lines PN and PQ , then $\cos \alpha$ is equal to _____.

- (A) $\frac{1}{\sqrt{5}}$
- (B) $\frac{\sqrt{3}}{2}$
- (C) $\frac{1}{\sqrt{3}}$
- (D) $\frac{1}{2\sqrt{3}}$

Question No. 5: (Only One Option Correct Type)

Box A contains 2 black and 3 red balls while box B contains 3 black and 4 red balls. Out of these two boxes one is selected at random and the probability of choosing box A is double that of box B . If a red ball is drawn from the selected box, then the probability that it has come from box B is

- (A) $\frac{21}{41}$
- (B) $\frac{10}{31}$
- (C) $\frac{12}{31}$
- (D) $\frac{13}{41}$

Question No. 6: (Only One Option Correct Type)

Let $I = \int \frac{(e^x - 1)(\sin x - \cos x) + x \cos x}{\sin^2 x + (e^x - 1 - x)^2} dx = \tan^{-1}(f(x)) + C$, where 'C' is the constant of integration and $\lim_{x \rightarrow 0} f(x) = 0$. If $\lim_{x \rightarrow 0} \frac{f(x)}{x} = \frac{p}{q}$, where $p, q \in N$ then find the least value of $(p + q)$.

- (A) 3
- (B) 2
- (C) 7
- (D) 5

Question No. 7: (Only One Option Correct Type)

The area of the region lying above X -axis, and included between the circle $x^2 + y^2 = 2ax$ and the parabola $y^2 = ax, a > 0$ is

- (A) $8\pi a^2$
- (B) $a^2 \left(\frac{\pi}{4} - \frac{2}{3} \right)$

(C) $\frac{16\pi a^2}{9}$

(D) $\pi \left(\frac{27}{8} + 3a^2 \right)$

Question No. 8: (Only One Option Correct Type)

$g(n) = \int_0^{n^2+n+1} e^{x/2-[x/2]} \left(\frac{x}{2} - \left[\frac{x}{2} \right] \right) d(x - [x]); n \in N$ then $g(n)$

(A) has minimum value as $\frac{1}{4} + \sqrt{e}$

(B) has maximum value as $3 - \sqrt{e}$

(C) has minimum value as $\frac{3}{4} - \sqrt{\frac{e}{4}}$

(D) none of these

Question No. 9: (Only One Option Correct Type)

A line passing through the point of intersection of $x + y = 4$ and $x - y = 2$ makes an angle $\tan^{-1} \left(\frac{3}{4} \right)$ with the X -axis. It intersects the parabola $y^2 = 4(x - 3)$ at points (x_1, y_1) and (x_2, y_2) , respectively. Then $|x_1 - x_2|$ is equal to .

(A) $\frac{16}{9}$

(B) $\frac{32}{9}$

(C) $\frac{40}{9}$

(D) $\frac{80}{9}$

Question No. 10: (Only One Option Correct Type)

A line L cuts the lines AB , AC and AD of a parallelogram $ABCD$ at points B_1 , C_1 and D_1 respectively. If

$\overrightarrow{AB_1} = \lambda_1 \overrightarrow{AB}$, $\overrightarrow{AD_1} = \lambda_2 \overrightarrow{AD}$ and $\overrightarrow{AC_1} = \lambda_3 \overrightarrow{AC}$ then $\frac{1}{\lambda_3} =$

(A) $\frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

(B) $\frac{1}{\lambda_1} - \frac{1}{\lambda_2}$

(C) $-\lambda_1 + \lambda_2$

(D) $\lambda_1 + \lambda_2$

Question No. 11: (Only One Option Correct Type)

For a positive integer m , if $\lim_{x \rightarrow \infty} \left(x^3 \ln \left(\frac{x+1}{x} \right) + \frac{x}{2} - x^2 \right) = \frac{1}{m}$. Then the value of m is

(A) 1

(B) 2

(C) 3

(D) 4

Question No. 12: (Only One Option Correct Type)

Let $f(x) = x^{2010} + x^{1010} - x^{510} + x^{210} + x^2$. If $f(x)$ is divided by $x^2(x^2 - 1)$, then we get remainder as $g(x)$, function of x . If $g(x)$ is defined from R to R , then:

(A) $g(x)$ is injective but not surjective

(B) $g(x)$ is surjective but not injective

(C) $g(x)$ is neither injective nor surjective

(D) $g(x)$ is injective and surjective

Question No. 13: (Only One Option Correct Type)

Let three positive numbers a, b, c are in geometric progression, such that $a, b + 8, c$ are in arithmetic progression and $a, b + 8, c + 64$ are in geometric progression. If the arithmetic mean of a, b, c is k , then $\frac{3}{13}k$ is equal to

(A) 4

(B) 8

(C) 6

(D) 2

Question No. 14: (Only One Option Correct Type)

The shortest distance between the lines $\vec{r} = (1 - t)\hat{i} + (t - 2)\hat{j} + (3 - 2t)\hat{k}$ and $\vec{r} = (p + 1)\hat{i} + (2p - 1)\hat{j} + (2p + 1)\hat{k}$ is

(A) $\frac{8}{\sqrt{29}}$ units

(B) $\frac{4}{\sqrt{29}}$ units

(C) $\frac{2}{\sqrt{5}}$ units

(D) $\frac{4}{\sqrt{19}}$ units

Question No. 15: (Only One Option Correct Type)

Consider the system of equations $2x + P^2y + 6z = 8$, $x + 2y + 2qz = 5$ and $x + y + 3z = 4$

(A) Given system has unique solution for $P \neq \pm\sqrt{2}$ and $q = \frac{3}{2}$

- (B) Given system has no solution for $P = \pm\sqrt{2}$ and $q = \frac{3}{2}$
- (C) Given system has infinite solution for $P = \pm\sqrt{2}$ and $q \in R$
- (D) None of these

Question No. 16: (Only One Option Correct Type)

Consider a differential equation

$$\left(x \tan\left(\frac{y}{x}\right) - y \sec^2\left(\frac{y}{x}\right)\right)dx + x \sec^2\left(\frac{y}{x}\right)dy = 0$$

with initial condition $y(2) = \frac{\pi}{4}$, then the value of $x \tan \frac{y}{x}$ is :

- (A) 0.82
- (B) 1.82
- (C) 2.65
- (D) 0.65

Question No. 17: (Only One Option Correct Type)

If the quadratic equation $x^2 + (2 - \tan \theta)x - (1 + \tan \theta) = 0$ has two integral roots, then sum of all possible values of θ in the interval $(0, 2\pi)$ is $k\pi$, then the value of k is equal to

- (A) 1
- (B) 4
- (C) 3
- (D) 2

Question No. 18: (Only One Option Correct Type)

Let $x_1, x_2, x_3 \dots x_k$ be k observations and $w_i = ax_i + b$ for $i = 1, 2, 3 \dots k$, where a and b are constants. If mean of x_i is 52 and their standard deviation is 12 and mean of w_i is 60 and their standard deviation is 15, then the value of a and b should be

- (A) $a = 1.25, b = -5$
- (B) $a = -1.25, b = 5$
- (C) $a = 2.5, b = -5$
- (D) $a = 2.5, b = 5$

Question No. 19: (Only One Option Correct Type)

Let $A(z_1)$, $B(z_2)$ and $C(z_3)$ lie on the circle $|z - i| = 1$ and satisfy the equation $3z_1 + i = 2z_2 + 2z_3$. If D is the centre of the circle $|z - i| = 1$, then area of quadrilateral $ABCD$ equals :-

- (A) $\frac{\sqrt{3}}{4}$
- (B) $\frac{\sqrt{5}}{4}$
- (C) $\frac{\sqrt{11}}{4}$
- (D) $\frac{\sqrt{7}}{4}$

Question No. 20: (Only One Option Correct Type)

The equation of the curve obtained by reflecting the ellipse $\frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1$ about the line $x - y - 2 = 0$ is $16x^2 + 9y^2 + k_1x - 36y + k_2 = 0$ then sum of prime factors of $(k_1 + k_2)$ is:-

- (A) 12
- (B) 9
- (C) 8
- (D) 16

Mathematics Numerical (Maximum Marks: 20)

Question No. 21: (Numerical Type)

Last two digits of the number 19^{94} is (for example, if last two digits are 06 report 6 and if last two digits are 23 report 23 as answer)

Question No. 22: (Numerical Type)

Let $A = \{x \mid x^3 + x^2 - px + q = 0, p, q, \in R\}$ and $B = \{x \mid x^2 - qx + 2 = 0, q \in R\}$ be the sets. If $n(A \cap B) = 2$ and $x_0 \in (A - B)$, then find the value of $|p - q + x_0|$.

Question No. 23: (Numerical Type)

Consider $\triangle ABC$, $A(5, -1)$, $B(\alpha, -7)$, $C(-2, \beta)$. Let $(-6, -4)$ is image of orthocentre of $\triangle ABC$ in the point mirror M which is mid-point of the side BC . Also (p, q) is circumcentre of triangle ABC , then the value of $\beta^2 - \alpha^2 + 5\beta - \alpha - 6p + 2q$ is _____.

Question No. 24: (Numerical Type)

Let $f : R - \{1\} \rightarrow R - \{1\}$ be a function satisfying the differential equation

$2x(y + x)dx - x^2(dx + dy) = (x + y)^2dx$ with $f(2) = 2$. If area enclosed by $y = f(x)$ and x -axis from $x = 2$ to $x = 3$ is $(a + \ln b)$ where $a, b \in \mathbb{N}$, then find the value of $(a + b)$.

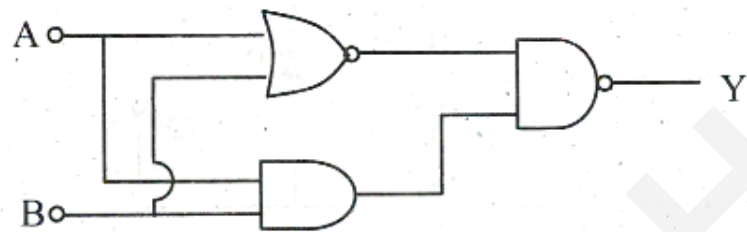
Question No. 25: (Numerical Type)

Let O be the interior point of $\triangle ABC$ such that $2\overrightarrow{OA} + 3\overrightarrow{OB} + 6\overrightarrow{OC} = \vec{0}$ where O is origin. If $\frac{\text{Area of } (\triangle ABC)}{\text{Area of } (\triangle AOB)} = \frac{m}{n}$, where m and n are relatively prime, then $(m - n)$ is equal to

Physics Single Correct (Maximum Marks: 80)

Question No. 26: (Only One Option Correct Type)

The truth table for the given logic circuit is



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

(P)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(Q)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

(R)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

(S)

- (A) P
- (B) Q
- (C) S
- (D) R

Question No. 27: (Only One Option Correct Type)

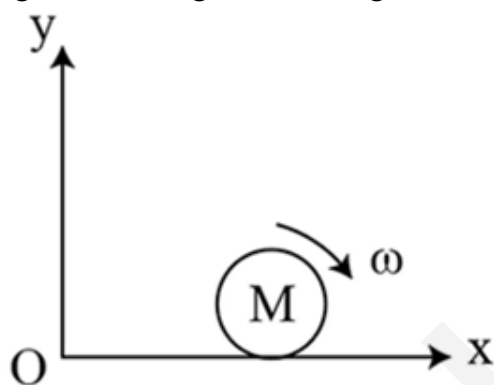
Assertion: When two soap bubbles having different radii are kept in contact, the common surface at their interface will bulge into a large bubble.

Reason: Pressure inside the smaller bubble is larger.

- (A) Assertion is true, reason is true, reason is correct explanation for assertion.
- (B) Assertion is true, reason is true, reason is not a correct explanation for assertion.
- (C) Assertion is true, reason is false.
- (D) Assertion is false, reason is true.

Question No. 28: (Only One Option Correct Type)

A disc of mass M and Radius R is rolling with an angular speed ω on the horizontal plane as shown in the figure. The magnitude of angular momentum of the disc about origin is :



- (A) $\frac{1}{2}MR^2\omega$
- (B) $MR^2\omega$
- (C) $\frac{3}{2}MR^2\omega$
- (D) $2MR^2\omega$

Question No. 29: (Only One Option Correct Type)

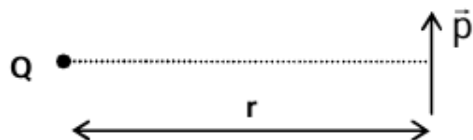
The radii of curvature of both the surfaces of a convex lens of focal length ' f ' and focal power ' P ' are equal. One of the surfaces is made plane by grinding. The new focal length and focal power of the lens is.

- (A) $\frac{2}{3}f, \frac{2}{3}P$
- (B) $\sqrt{\frac{2}{f}}, \sqrt{\frac{P}{2}}$
- (C) $\frac{f}{2}, 2P$

(D) $2f, \frac{P}{2}$

Question No. 30: (Only One Option Correct Type)

An infinitesimal electric dipole of dipole moment $\vec{p}$ is placed at a distance r from a point positive charge Q such that the direction of dipole moment $\vec{p}$ is perpendicular to the line joining the point charge Q and centre of the dipole. If the magnitudes of force and torque exerted by the point charge Q on the dipole are F and τ respectively, then



(A) $F = \frac{3Qp}{4\pi\epsilon_0 r^3}$

(B) $F = \frac{Qp}{2\pi\epsilon_0 r^3}$

(C) $\tau = \frac{Qp}{2\pi\epsilon_0 r^2}$

(D) $\tau = \frac{Qp}{4\pi\epsilon_0 r^2}$

Question No. 31: (Only One Option Correct Type)

Consider a glass slab which is silvered at one side and the other side is transparent. Given the refractive index of the glass slab to be 1.5. If a ray of light is incident at an angle of 45° on the transparent side, then the deviation of the ray of light from its initial path, when it comes out of the slab is

(A) 180°

(B) 120°

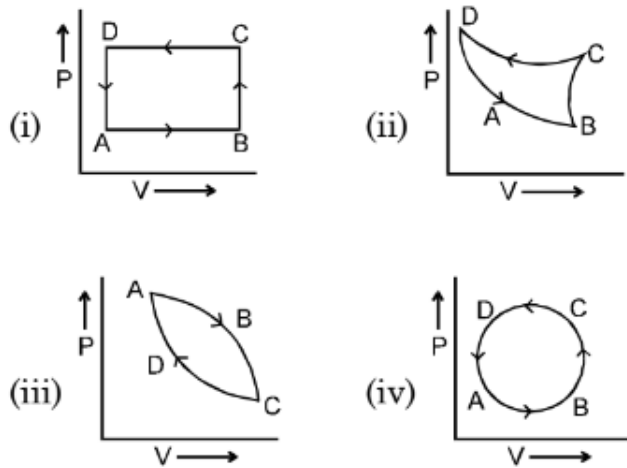
(C) 45°

(D) 90°

Question No. 32: (Only One Option Correct Type)

In the following figures (i) to (iv), variation of volume by change of pressure is shown. A gas is taken along the

path ABCDA. The change in internal energy of the gas will be:



- (A) positive in all cases from (i) to (iv)
- (B) positive in cases (i), (ii) and (iii) but zero in case (iv)
- (C) negative in cases (i), (ii) and (iii) but zero in case (iv)
- (D) zero in all the four cases.

Question No. 33: (Only One Option Correct Type)

A straight wire carrying current i is turned into a circular loop. If the magnitude of magnetic moment associated with it in MKS units is M , the length of wire will be?

- (A) $4\pi i M$
- (B) $\sqrt{\frac{4\pi M}{i}}$
- (C) $\sqrt{\frac{4\pi i}{M}}$
- (D) $\frac{M\pi}{4i}$

Question No. 34: (Only One Option Correct Type)

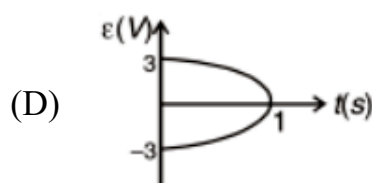
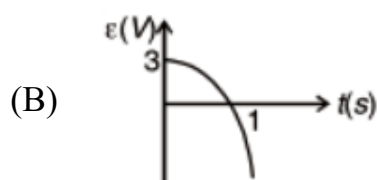
Consider the following statements about interference of light

- A. The interference fringes are equally bright and equally spaced
 - B. At the centre of a bright fringe, the intensity is four times the intensity of the incident wave
 - C. For constructive interference of two waves, the crest of one wave coincides with trough of another wave.
- Which of the above statements are correct?

- (A) A and B only
- (B) A and C only
- (C) All A, B and C
- (D) B and C only

Question No. 35: (Only One Option Correct Type)

The magnetic flux linked with a circuit is given by $\phi = t^3 - 3t - 7$. Choose the correct graph between induced emf $[\varepsilon]$ and time $[t]$.



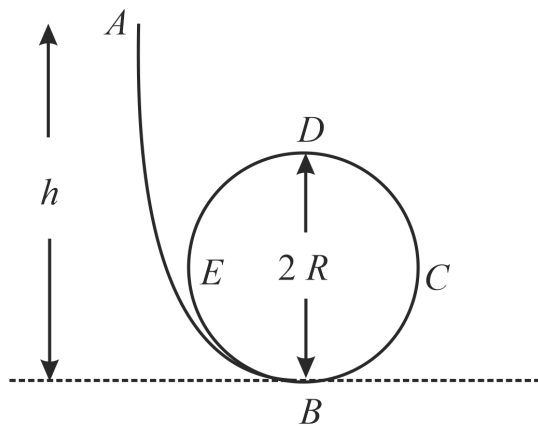
Question No. 36: (Only One Option Correct Type)

A ball is projected with a certain speed from a point on the ground which is at a distance of 30 m from a vertical wall. If the angle of projection is 45° with the horizontal, the ball just clears the top of the wall and strikes the ground at a distance of 10 m from the wall on the other side. The height of wall is h . Find $10h$. (Take $g = 10 \text{ ms}^{-2}$)

- (A) 125
- (B) 50
- (C) 100
- (D) 75

Question No. 37: (Only One Option Correct Type)

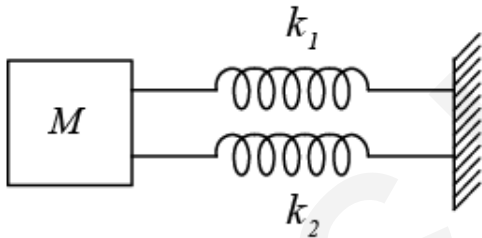
A frictionless track $ABCDE$ ends in a circular loop of radius R . A body slides down the track from the point A which is at a height $h = 5 \text{ cm}$. Maximum value of R for the body to successfully complete the loop is



- (A) 5 cm
- (B) $\frac{15}{4}$ cm
- (C) $\frac{10}{3}$ cm
- (D) 2 cm

Question No. 38: (Only One Option Correct Type)

Two springs with negligible masses and force constants $k_1 = 200 \text{ N/m}$ and $k_2 = 160 \text{ N/m}$ are attached to the block of mass $m = 10 \text{ kg}$ as shown in figure. Initially the block is at rest, at the equilibrium position in which both springs are neither stretched nor compressed. At time $t = 0$, sharp impulse of 50 N-s is given to the block with a hammer along the spring. Then the amplitude of spring will be K metre. Find $6K$



- (A) 8
- (B) 5
- (C) 4
- (D) 6

Question No. 39: (Only One Option Correct Type)

An inductor of reactance 1Ω and a resistor of resistance 3Ω are connected in series to the terminals of 10 V (rms) AC source. The power dissipated in the circuit is -

- (A) 25
- (B) 30
- (C) 35

(D) 20

Question No. 40: (Only One Option Correct Type)

For the equation $F = A^a v^b d^c$, where F is the force, A is the area, v is the velocity and d is the density, the values of a , b and c are respectively

(A) 1,2,1

(B) 2,1,1

(C) 1,1,2

(D) 0,1,1

Question No. 41: (Only One Option Correct Type)

Ethanol of density $\rho = 700 \text{ kg/m}^3$ flows smoothly through a horizontal pipe that tapers in crosssectional area from $A_1 = 1.2 \times 10^{-3} \text{ m}^2$ to $A_2 = \frac{A_1}{2}$. The pressure difference between the wide and the narrow sections of pipe is 4200 Pa. What is the volume flow rate of ethanol in multiples of $10^{-4} \text{ m}^3/\text{s}$.

(A) 12

(B) 16

(C) 24

(D) None of these

Question No. 42: (Only One Option Correct Type)

Two identical photons of energy $E(4\text{eV} \leq E \leq 8\text{eV})$ fall on two different metals whose work functions are in the ratio of 1 : 2. The ratio of the kinetic energies of the most energetic electrons coming from each metal is 2 : 1. Work function of one of the metal is 4 eV . What is the energy of photon in eV .

(A) 6 eV

(B) 8 eV

(C) 4 eV

(D) 5 eV

Question No. 43: (Only One Option Correct Type)

A cylindrical wire has a mass $(0.3 \pm 0.003)\text{g}$, radius $(0.5 \pm 0.005)\text{mm}$ and length $(6 \pm 0.06)\text{cm}$. The maximum percentage error in the measurement of its density is -

(A) 1

- (B) 2
- (C) 3
- (D) 4

Question No. 44: (Only One Option Correct Type)

A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T . Neglecting all vibrational modes, the total internal energy of the system is (R is universal gas constant)

- (A) $4 RT$
- (B) $15 RT$
- (C) $9 RT$
- (D) $11 RT$

Question No. 45: (Only One Option Correct Type)

The position (x) of a body moving along x-axis at time (t) is given by $x = 3t^2$, where x is in metre and t is in second. If mass of body is 2 kg, then find the instantaneous power delivered to body by force acting on it at $t = 4$ s : –

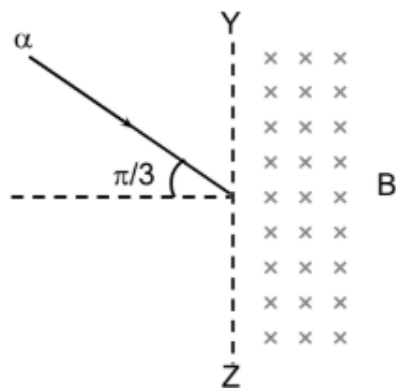
- (A) 288 W
- (B) 280 W
- (C) 290 W
- (D) 260 W

Physics Numerical (Maximum Marks: 20)

Question No. 46: (Numerical Type)

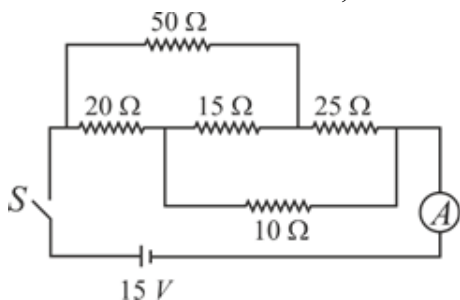
A particle with specific charge α , enters a magnetic field of magnitude B , existing only to the right of the boundary YZ , as shown. The direction of motion of the particle is perpendicular to the direction of B . If the

time spent by the particle in the field is $K \times \frac{\pi}{3\alpha B}$, find the value of 'K'



Question No. 47: (Numerical Type)

When switch S is closed, then the reading of ammeter is A , then value of $10A$ is _____



Question No. 48: (Numerical Type)

If a slab of insulating material, 4×10^{-3} m thick is introduced between the plates of a parallel plate capacitor, the separation between the plates has to be increased by 3.5×10^{-3} m to restore the capacity to original value. The dielectric constant of the material will be

Question No. 49: (Numerical Type)

Two solid balls have different radii but are made of same material. The balls are linked together with a long thin thread and released from a large height. At the terminal velocity, the thread is under tension. The larger ball has a fixed mass, but we have choice of the smaller ball with different masses. At what ratio of larger and smaller mass will this tension be maximum?

Question No. 50: (Numerical Type)

If B_1 is the magnetic field induction at a point on the axis of a circular coil of radius R situated at a distance $R\sqrt{3}$ and B_2 is the magnetic field at the centre of the coil, then the ratio of $\frac{B_1}{B_2}$ is equal to K, find 16K.

Chemistry Single Correct (Maximum Marks: 80)

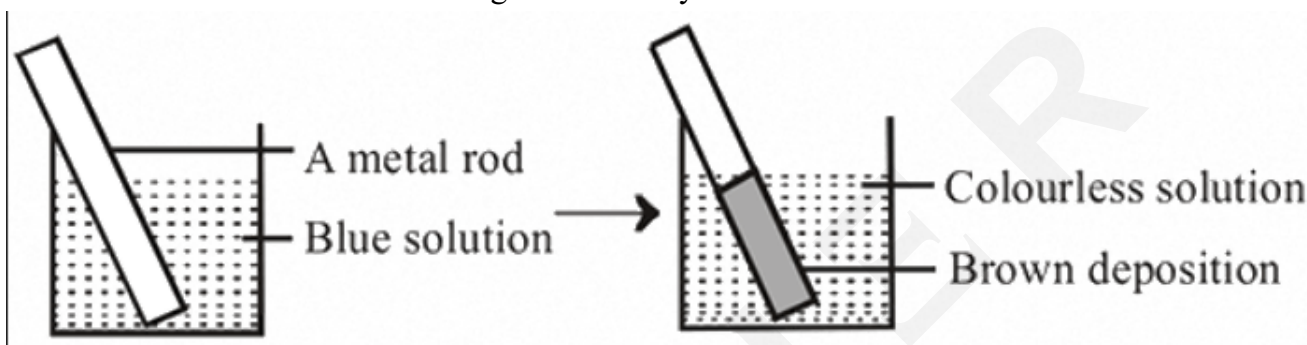
Question No. 51: (Only One Option Correct Type)

Dissolving 120 g of a compound (mol. wt = 60) in 1000 g of water gave a solution of density 1.12 g mL^{-1} . The molarity of solution is ?

- (A) 1.5
(B) 2
(C) 3
(D) 3.5

Question No. 52: (Only One Option Correct Type)

A redox reaction is shown in the diagrams. Identify the reaction.



- (A) $\text{Zn}(s) + \text{Cu}^{2+}(aq) \longrightarrow \text{Zn}^{2+}(aq) + \text{Cu}(s)$
(B) $\text{Cu}(s) + 2\text{Ag}^{+}(aq) \longrightarrow \text{Cu}^{2+}(aq) + 2\text{Ag}(s)$
(C) $2\text{Ag}(s) + \text{Cu}^{2+}(aq) \longrightarrow 2\text{Ag}^{+}(aq) + \text{Cu}(s)$
(D) $\text{Cu}(s) + \text{Zn}^{2+}(aq) \longrightarrow \text{Cu}^{2+}(aq) + \text{Zn}(s)$

Question No. 53: (Only One Option Correct Type)

Match List - I with List - II.

List-I (complexes)		List-II (magnetic moment)	
(a)	$[\text{Fe}(\text{CN})_6]^{3-}$	(i)	5.92 BM
(b)	$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	(ii)	0 BM
(c)	$[\text{Fe}(\text{CN})_6]^{4-}$	(iii)	4.90 BM
(d)	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(iv)	1.73 BM

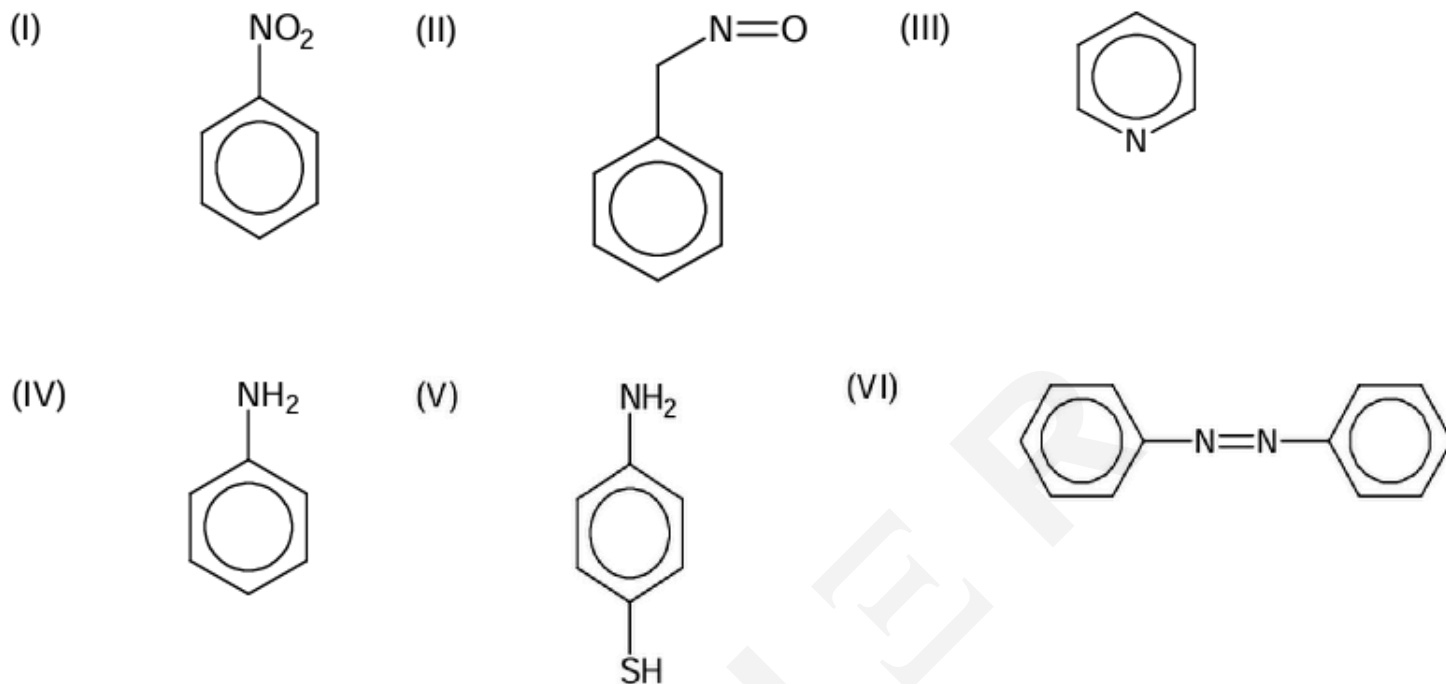
Choose the correct answer from the options given below.

- (A) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
(B) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
(C) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(D) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

Question No. 54: (Only One Option Correct Type)

Consider the following compounds:



Let, the compound(s) which cannot be Kjeldahlised be ' x ' the compound(s) which gives blue colour in Lassaigne's test of nitrogen be y and the compound(s) which gives red colour in Lassaigne's test of nitrogen be z . So, the value of $\left(\frac{x+y}{z}\right)$ will be:

- (A) 4
- (B) 5
- (C) 9
- (D) 10

Question No. 55: (Only One Option Correct Type)

Two reactions, $A \rightarrow \text{Products}$ and $B \rightarrow \text{Products}$, have rate constants K_a and K_b at temperature T and activation energies E_a and E_b respectively. If $K_a > K_b$ and $E_a < E_b$ and assuming that A for both the reactions is same then :

- (A) At higher temperatures K_a will be lesser than K_b
- (B) At lower temperature K_a and K_b will differ more and $K_a > K_b$
- (C) As temperature rises K_a and K_b will be close to each other in magnitude
- (D) None of these

Question No. 56: (Only One Option Correct Type)

What are X and Y respectively in the following reaction?



- (A) Na/NH_3 (liq.) and $\text{Pd}/\text{BaSO}_4 + \text{H}_2$
- (B) $\text{Ni}/140^\circ\text{C}$ and $\text{Pd}/\text{BaSO}_4 + \text{H}_2$
- (C) $\text{Ni}/140^\circ\text{C}$ and Na/NH_3 (liq.)
- (D) $\text{Pd}/\text{BaSO}_4 + \text{H}_2$ and Na/NH_3 (liq.)

Question No. 57: (Only One Option Correct Type)

Which of the following is a disproportionation reaction?

- (A) $\text{Cl}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{ClO}^-(\text{aq}) + \text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- (B) $\text{Cl}_2(\text{g}) + 2\text{I}^-(\text{aq}) \rightarrow 2\text{Cl}^-(\text{aq}) + \text{I}_2(\text{s})$
- (C) $2\text{Fe}(\text{s}) + 3\text{H}_2\text{O}(\text{l}) \xrightarrow{\Delta} \text{Fe}_2\text{O}_3(\text{s}) + 3\text{H}_2(\text{g})$
- (D) $2\text{H}_2\text{O}(\text{l}) + 2\text{F}_2(\text{g}) \rightarrow 4\text{HF}(\text{aq}) + \text{O}_2(\text{g})$

Question No. 58: (Only One Option Correct Type)

Zirconium phosphate $[\text{Zr}_3(\text{PO}_4)_4]$ dissociates into three zirconium cations of charge +4 and four phosphate anions of charge -3. If molar solubility of zirconium phosphate is denoted by S and its solubility product by K_{sp} then which of the following relationship between S and K_{sp} is correct ?

- (A) $S = \{K_{\text{sp}}/6912\}^{1/7}$
- (B) $S = \{K_{\text{sp}}/144\}^{1/7}$
- (C) $S = \left\{K_{\text{sp}}/(6912)^{1/7}\right\}$
- (D) $S = (K_{\text{sp}}/6912)^{1/7}$

Question No. 59: (Only One Option Correct Type)

Select the set having incorrect statements given here.

- (i) Manganese exhibits +7 oxidation state
- (ii) Zinc forms coloured ions
- (iii) $[\text{CoF}_6]^{3-}$ is diamagnetic

(iv) Sc forms +4 oxidation state

(v) Zn exhibits only +2 oxidation state

(A) (i), (ii) and (iii)

(B) (ii), (iii) and (iv)

(C) (i), (iii) and (iv)

(D) (i), (iii) and (v)

Question No. 60: (Only One Option Correct Type)

An ideal gas is expanded from (p_1, V_1, T_1) to (p_2, V_2, T_2) under different conditions. The incorrect statement among the following is?

(A) The work done by the gas is less when it is expanded reversibly from V_1 to V_2 under adiabatic conditions as compared to that when expanded reversibly from V_1 to V_2 under isothermal conditions.

(B) The change in internal energy of the gas is (i) zero, if it is expanded reversibly with $T_1 = T_2$, and (ii) positive, if it is expanded reversibly under adiabatic conditions with $T_1 \neq T_2$.

(C) If the expansion is carried out freely, it is simultaneously both isothermal as well as adiabatic

(D) The work done on the gas is maximum when it is compressed irreversibly from (p_2, V_2) to (p_1, V_1) against constant pressure P_1 .

Question No. 61: (Only One Option Correct Type)

A tetrapeptide has $-\text{COOH}$ group on alanine. This produces glycine (Gly), valine (Val), phenyl alanine (Phe) and alanine (Ala), on complete hydrolysis. For this tetrapeptide, the number of possible sequences (primary structures) with $-\text{NH}_2$ group attached to a chiral center is-

(A) 2

(B) 3

(C) 4

(D) 5

Question No. 62: (Only One Option Correct Type)

If photons of energy 12.75 eV are passing through hydrogen gas in ground state then no. of lines in emission spectrum will be

(A) 6

- (B) 4
(C) 3
(D) 2

Question No. 63: (Only One Option Correct Type)

Assertion : More is the electron affinity greater is the reducing character.

Reason : Reducing character depends on oxidation potential.

- (A) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(B) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(C) If assertion is true but reason is false.
(D) If assertion is false but reason is true.

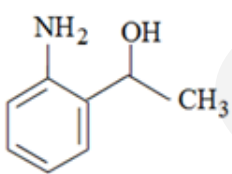
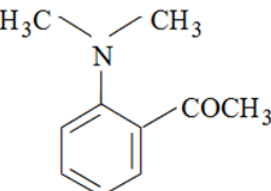
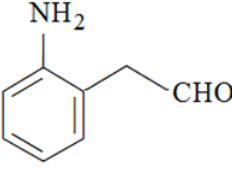
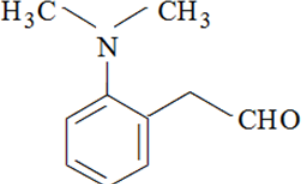
Question No. 64: (Only One Option Correct Type)

The tests performed on compound X and their inferences are

Test Inference

- (1) 2,4- DNP test coloured precipitate
(2) iodoform test yellow precipitate
(3) Azo dye test no dye formation

Compound 'X' is

- (A) 
- (B) 
- (C) 
- (D) 

Question No. 65: (Only One Option Correct Type)

How many isomeric dienes with a six membered ring are possible of the compound with the molecular formula C_7H_{10} ?

- (A) 6
- (B) 3
- (C) 5
- (D) 7

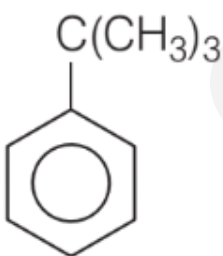
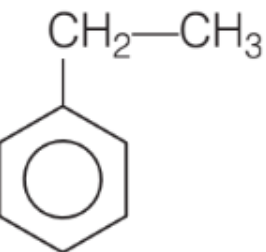
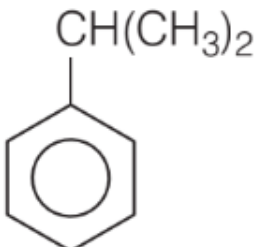
Question No. 66: (Only One Option Correct Type)

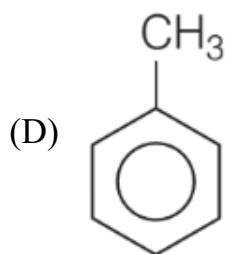
The reaction condition leading to the best yield of C_2H_5Cl are?

- (A) C_2H_6 (excess) + $Cl_2 \xrightarrow{\text{UV light}}$
- (B) $C_2H_6 + Cl_2$ (excess) $\xrightarrow[\text{room temp.}]{\text{dark}}$
- (C) $C_2H_6 + Cl_2 \xrightarrow{\text{UV light}}$
- (D) $C_2H_6 + Cl_2$ (excess) $\xrightarrow{\text{UV light}}$

Question No. 67: (Only One Option Correct Type)

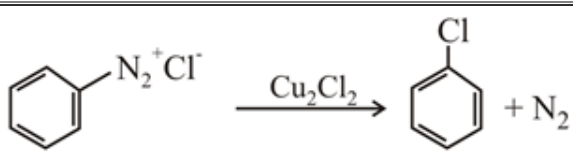
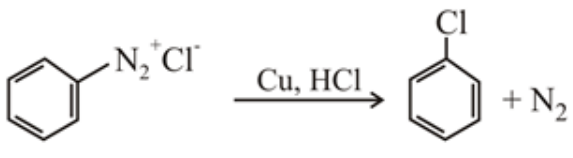
Identify the compound, which has maximum number of no bond resonance structures.

- (A) 
- (B) 
- (C) 



Question No. 68: (Only One Option Correct Type)

Match List-I with List-II.

List-I		List-II	
(a)		(i)	Wurtz reaction
(b)		(ii)	Sandmeyer reaction
(c)	$2 \text{CH}_3 \text{CH}_2 \text{Cl} + 2\text{Na} \xrightarrow{\text{Ether}} \text{C}_2\text{H}_5 - \text{C}_2\text{H}_5 + 2 \text{NaCl}$	(iii)	Fittig reaction
(d)	$2\text{C}_6\text{H}_5\text{Cl} + 2\text{Na} \xrightarrow{\text{Ether}} \text{C}_6\text{H}_5 - \text{C}_6\text{H}_5 + 2 \text{NaCl}$	(iv)	Gatterman reaction

Choose the correct answer from the options given below:

- (A) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (i), (d) $\rightarrow$ (ii)
- (B) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (iii)
- (C) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (ii)
- (D) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (i), (d) $\rightarrow$ (iii)

Question No. 69: (Only One Option Correct Type)

Which of the following molecules does not exhibit dipole moment? (i) CCl_4 (ii) CO_2 (iii) NH_3 (iv) CHCl_3 (v) H_2O (vi) $\text{CH}_3 - \text{O} - \text{CH}_3$

- (A) (ii), (v), (iv)
- (B) (i), (iii), (vi)
- (C) (i), (ii)
- (D) (iii), (iv), (vi)

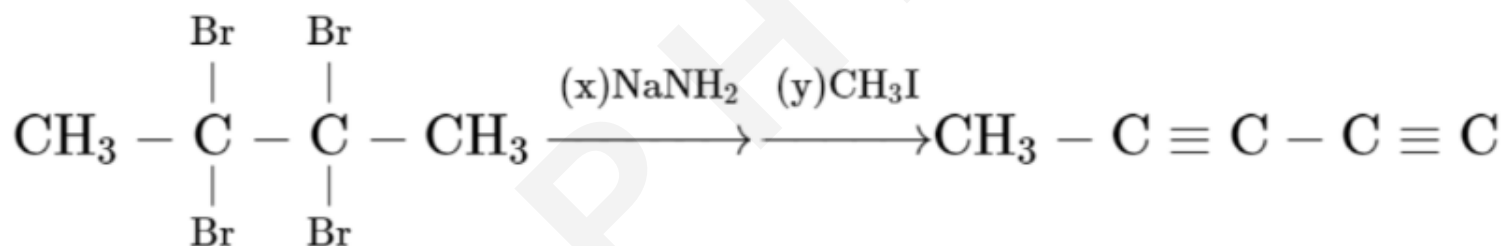
Question No. 70: (Only One Option Correct Type)

Which of the following is not correctly matched?

- (A) Acidic oxides P_2O_5 , NO_2 , Cl_2O_7
- (B) Basic oxides Na_2O , CaO , MgO
- (C) Neutral oxides CO_2 , CO , BeO
- (D) Amphoteric oxides ZnO , SnO , Al_2O_3

Chemistry Numerical (Maximum Marks: 20)

Question No. 71: (Numerical Type)



x and y mole consumed. Find the value of $x + y$?

Question No. 72: (Numerical Type)

A 0.50 g mixture of Cu_2O and CuO contains 0.425 g of Cu. Mass of CuO (in g) in mixture is K, find value of 10 K in the mixture? (Cu = 63.5 u, O = 16 u)

(Mark answer to nearest integer)

Question No. 73: (Numerical Type)

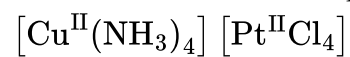
The bond dissociation energies for Cl_2 , I_2 and ICl are 242.3, 151 and 211.3 kJ/mol respectively. The enthalpy of sublimation of iodine is 62.8 kJ/mol. What is the standard enthalpy of formation of ICl (g) ? (roundoff answer to nearest integer in KJ/mol)

Question No. 74: (Numerical Type)

Among the elements of lanthanoid series (${}_{58}\text{Ce}$ – ${}_{71}\text{Lu}$), the total number of elements which have odd number of electrons in f-orbitals in their ground state configuration

Question No. 75: (Numerical Type)

Find the total number of possible isomers for the complex compound



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